Write a C++ program to implement solution of producer-consumer problem. The program reads size of buffer from command line argument and uses semaphore as synchronization primitive. Producer is producing a non-decreasing in integer sequence. The program should correctly handle synchronization and avoid: Race conditions, Buffer overflow, Buffer underflow and Deadlock.

1) Implement the solution where the producer and consumer execute as threads of the same process.

2) Implement the solution where the producer and consumer are independent processes.

3) Implement solution for multiple producer and consumer threads. Buffer size, number of producers and number of consumers are passed vide command line arguments. Here each producer shall output item as p.x where p is producer Id and X is integer.

Every time producer/consumer thread is scheduled, it should produce message of item X produced/consumed where X is an integer produced by producer.

The program must ensure that:

- Multiple producers can safely access the buffer.
- Multiple consumers can safely access the buffer.
- No producer inserts an item when the buffer is full.
- No consumer removes an item when the buffer is empty.
- Each item is consumed exactly once.
- Access to the shared buffer is mutually exclusive.